Kavi Ishwarrya S K

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LinkedIn | GitHub | Portfolio | LeetCode

SUMMARY

B.E. CSE (Internet of Things) student (graduating 2027) focused on embedded systems. Built a two-node ESP32 monitoring system over CAN bus on FreeRTOS, with my own I2C sensor driver, node-loss detection and cloud upload. Hands-on PCB design experience in KiCad from an internship, and 250+ LeetCode problems solved in Java.

EDUCATION

Nandha Engineering College, Erode, Tamil Nadu 2023 to 2027
B.E. in Computer Science and Engineering (Internet of Things) CGPA: 8.3/10

EXPERIENCE

Elite Systems and Controls Dec 2025 to Jan 2026
Intern, PCB and Embedded Design

- Designed circuit schematics and **8 PCB layouts** in the **KiCad** PCB Editor.
- Assisted with a 3-stage PCB design workflow covering schematic capture, layout optimisation and design validation.

Nandha InfoTech Jun 2025 to Jul 2025
Trainee, IoT Systems

- Built a **smart home automation system in Embedded C/C++** with an **Arduino**, **ESP8266** Wi-Fi module and **DHT11** sensor.
- Linked it to the **Blynk IoT platform** for remote temperature and humidity control; tested and documented it.

PROJECTS

EnviroNode: Two-Node Environmental Monitoring over CAN Bus github.com/kaviish123/environode-esp32-can
ESP32, FreeRTOS, CAN (TWAI), I2C, UART, Wi-Fi, ThingSpeak

- Built a **two-node ESP32 system**: a sensor node sends readings over **CAN bus** to a gateway; a heartbeat flags a lost node in 3 s.
- Wrote my own **register-level I2C driver** for the **BMP280** from the Bosch datasheet, including the calibration formulas.
- Gateway shows live data on an **I2C OLED** and uploads to **ThingSpeak** over Wi-Fi; verified 120+ CAN frames with zero loss.
- Split the firmware into **FreeRTOS tasks** with mutexes on shared data, and wrote test sketches to debug wiring faults.

RideBoard, Ride-Sharing Mobile App github.com/kaviish123/RideBoardApp
React Native, Expo, Firebase Authentication, Cloud Firestore

- Built a ride-sharing app for **iOS and Android** with **React Native** and **Expo**, using **Firebase Authentication** for sign-in.
- Wrote the features to post, browse, edit and delete ride listings, with real-time storage in **Cloud Firestore**.
- Kept the UI modular and responsive, used **Git/GitHub** for version control and made Android builds with Expo EAS Build.

AWS Static Website Hosting github.com/kaviish123/AWS-Static-Website
AWS, Amazon S3

- Hosted a static website (HTML, CSS, JavaScript) on **Amazon S3** with bucket permissions and public access set up.

RESEARCH

Sales Forecasting Using Customer Segments and Holiday Effects *Unpublished manuscript*
First author, with Karthik V, Keerthana K and Maheswari S (faculty guide), Nandha Engineering College

- Used **LightGBM** ensemble learning on real retail sales data to study how customer segments and holidays affect sales.

TECHNICAL SKILLS

Embedded Systems: ESP32, Arduino, ESP8266, FreeRTOS, Embedded C/C++, KiCad (schematic and PCB design)
Protocols & Interfaces: I2C, UART, CAN (ESP32 TWAI), Wi-Fi, HTTP; BMP280, SSD1306 OLED, SN65HVD230
Programming & Databases: C/C++, Java, Python, SQL, MySQL
Cloud & Mobile: AWS (S3, EC2, IAM, static website hosting), ThingSpeak, React Native, Firebase
Core CS & Tools: DBMS, Operating Systems, Computer Networks, OOP; Git, GitHub, VS Code, AutoCAD, Excel

CERTIFICATIONS & TRAINING

- AWS Cloud Practitioner Essentials, AWS (2025); Cloud Computing, NPTEL, IIT Kharagpur, 12-week course (2024)
- Security Engineering for the IoT, Infosys Springboard (2025); Introduction to IoT and Digital Transformation, Cisco (2025)
- Java Programming for Beginners, Simplilearn (2026); Infosys Springboard Pragati: Path to Future, Cohort 3, 11 weeks (2024 to 2025)

ACHIEVEMENTS & EXTRACURRICULAR

- **2nd Runner-up, Language Geek** (technical event), SAMHITA '24 national symposium, MIT, Anna University, Chennai (Mar 2024)
- Exhibited a project, **Live Portfolio Website**, at the Project Expo, Innovation Day '24, Nandha Engineering College (Feb 2024)
- Solved **250+ problems on LeetCode** in Java covering data structures and algorithms, with 50 Days and 100 Days badges (2026)
- Took part in Project Presentation and Blind Coding at GUSTO 2024, Government College of Engineering, Erode (Apr 2024)
- **Workshops:** IoT Sensors (Nandha), AI and Python (Mechanica, IIT Madras Research Park), Generative AI (Skill Radar), Data Science (MIT)